File permissions in Linux

Project description

In this project, I used Linux commands to view, interpret and modify file and directory permission. I learned how to inspect permissions, understand permission strings, and used the chmod command to update access rights for files, hidden files and directories.

Check file and directory details

I used the pwd, cd, ls -l, and ls -la commands to navigate directories and view file and directory details. The ls -la command displayed all files, including hidden files, along with their permissions, owner, group, file size, and modification date. This allowed me to review the current permissions before making any changes.

```
researcher2@90f232a0ba9b:~$ pwd
/home/researcher2
researcher2@90f232a0ba9b:~$ cd projects
researcher2@90f232a0ba9b:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Aug  7 06:16 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Aug  7 06:16 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Aug  7 06:16 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Aug  7 06:16 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Aug  7 06:16 project_t.txt
researcher2@90f232a0ba9b:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Aug  7 06:16 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug  7 06:49 ..
-rw--w---- 1 researcher2 research_team   46 Aug  7 06:16 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Aug  7 06:16 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Aug  7 06:16 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Aug  7 06:16 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Aug  7 06:16 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Aug  7 06:16 project_t.txt
researcher2@90f232a0ba9b:~/projects$
```

Describe the permissions string

The permissions string consists of 10 characters. The first character indicates whether the item is a file (-) or a directory (d). The remaining nine characters are divided into three groups that represent the permissions for the user, group, and others. Each group contains r (read), w (write), and x (execute), while a dash (-) indicates that a permission is not granted.

.project_x.txt → -rw- -w - - - -
drafts → drwx- -x- - -
project_k.txt → -rw-rw-rw-
project_m.txt → -rw-r- - - - -
project_r.txt → -rw-rw-r- -
project_t.txt → -rw-rw-r- -

Change file permissions

I used the chmod to change file permissions by adding or removing read, write and execute access. This helped ensure that only authorised users could access or modify files.

```
researcher2@40d6b9f3ebd2:~/projects$ chmod o-w project_k.txt
researcher2@40d6b9f3ebd2:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Aug  7 04:54 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Aug  7 04:54 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Aug  7 04:54 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Aug  7 04:54 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Aug  7 04:54 project_t.txt
researcher2@40d6b9f3ebd2:~/projects$ chmod g-r project_m.txt
researcher2@40d6b9f3ebd2:~/projects$ 
```

Change file permissions on a hidden file

I used the chmod command to modify the permissions of a hidden file. I verified the changes using ls -la which displays hidden files along with their permission settings.

```
researcher2@40d6b9f3ebd2:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Aug  7 04:54 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug  7 05:21 ..
-rw--w---- 1 researcher2 research_team   46 Aug  7 04:54 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Aug  7 04:54 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Aug  7 04:54 project_k.txt
-rw------- 1 researcher2 research_team   46 Aug  7 04:54 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Aug  7 04:54 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Aug  7 04:54 project_t.txt
researcher2@40d6b9f3ebd2:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@40d6b9f3ebd2:~/projects$
```

Change directory permissions

I updated directory permissions using chmod command to control who could access, view, or modify the contents of the directory. I confirmed the permission changes by listing the directory details.

```
researcher2@40d6b9f3ebd2:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Aug  7 04:54 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Aug  7 04:54 project_k.txt
-rw------- 1 researcher2 research_team   46 Aug  7 04:54 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Aug  7 04:54 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Aug  7 04:54 project_t.txt
researcher2@40d6b9f3ebd2:~/projects$ chmod g-x drafts
researcher2@40d6b9f3ebd2:~/projects$
```

Summary

This project strengthened my understanding of Linux file permissions and access control. I learned how to inspect permission settings, interpret permission strings, and securely manage permissions for files, hidden files, and directories using Linux commands.